

MATHEMATICS

TARGET : JEE- Advanced 2023

CAPS-18

Area Under the Curves

SCQ (Single Correct Type) :

- Let $f(x) = 2\sqrt{x}$ and $g(x) = 2\sqrt{1-x}$ be two functions then the area bounded by $y = f(x)$, $y = g(x)$ and x axis is
 (A) $\frac{1}{3\sqrt{2}}$ sq unit (B) $\frac{2}{3\sqrt{2}}$ sq unit (C) $\frac{1}{\sqrt{2}}$ sq unit (D) $\frac{4}{3\sqrt{2}}$ sq unit
- The area bounded by the lines $y = 2$, $x = 1$, $x = a$ and the curve $y = f(x)$, which cuts the last two lines above the first line for all $a \geq 1$, is equal to $\frac{2}{3}((2a)^{3/2} - 3a + 3 - 2\sqrt{2})$ then $f(x) =$
 (A) $2\sqrt{2x}$; $x \geq 1$ (B) $\sqrt{2x}$; $x \geq 1$ (C) $2\sqrt{x}$; $x \geq 1$ (D) None of these
- A variable circle C touches $y = 0$ and passes through the point $(0,1)$. Let the locus of the centre of C is the curve P . The area enclosed by the curve P and the line $x + y = 2$ is
 (A) $\frac{14}{3}$ (B) $\frac{16}{3}$ (C) $\frac{15}{3}$ (D) $\frac{11}{3}$
- Let $n S_n$ be the area of the figure enclosed by a curve $y = x^3(1-x^2)^n$, $0 \leq x \leq 1$ and the x -axis. The value of $\lim_{n \rightarrow \infty} \sum_{k=1}^n S_k$ is
 (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) 1
- Find the area of the region enclosed between the curve $y = 2x^4 - x^2$, the x -axis and the ordinates of the points where the curve has local minima
 (A) $\frac{7}{60}$ (B) $\frac{7}{80}$ (C) $\frac{7}{120}$ (D) $\frac{7}{160}$
- Let $f(x) = \begin{cases} \ln(x - [x]); & x \notin I \\ 0; & x \in I \end{cases}$ Area enclosed by curves $y = f(x)$, $y = f(|x|)$, $-5 \leq x \leq 5$ and $y = 0$ is
 (A) $5(1 - \ln 2)$ (B) 0 (C) $5 \ln 2$ (D) none of these
- The area bounded by the curves $y = 2 - |x - 1|$, $y = \sin x$; $x = 0$ and $x = 2$ is _____.
 (A) $1 + 2\cos^2 1$ (B) $2 + \sin^2 1$ (C) $\frac{\pi}{2}$ (D) $1 + \log 2$
- The area bound by the curve $y = f(x)$, the x -axis and the line $y = 1$, where $f(x) = 1 + \frac{1}{x} \int_1^x f(t) dt$, ($x \neq 0$) is _____.
 (A) $2\left(1 - \frac{1}{e}\right)$ (B) $2\left(1 + \frac{1}{e}\right)$ (C) $\left(1 + \frac{1}{e}\right)$ (D) $\left(1 - \frac{1}{e}\right)$

9. Given $f(x) = x^3 + x - 2$, $f: \mathbb{R} \rightarrow \mathbb{R}$, area of region bounded by $y = f(x)$, $y = f^{-1}(x)$ and $xy = 0$ in 1st quadrant is A , where $A = 3t - \frac{5}{2}$, where $t = a^{1/b}$ and $3(a+b) = n$; $a, b, n \in \mathbb{N}$ and a is prime. Difference of last two digits of a^{11-36} is
- (A) 0 (B) 1 (C) 2 (D) 3

MCQ (One or more than one correct) :

10. The area bounded by the curve $y = f^{-1}(x)$, x -axis, $x = 0$ and $x = \pi$, where $f(x)$ is given by $f(x) = x + \sin x$ is $\frac{\pi^2 - a}{b}$ then the correct statement are :
- (A) $a = 2b$ (B) $a + 2b = 8$ (C) $3a = b$ (d) $\left(\frac{2\pi}{b}, \frac{a\pi}{4}\right)$ lies on $f(x)$
11. If A_i is the area bounded by $|x - a_i| + |y| = b_i$, $i \in \mathbb{N}$ where $1 \leq a_{i+1} = a_i + \frac{3}{2}b_i$ and $b_{i+1} = \frac{b_i}{2}$, $a_1 = 0$, $b_1 = 32$, then ____.
- (A) $A_3 = 128$ (B) $A_3 = 256$ (C) $\lim_{n \rightarrow \infty} \sum_{i=1}^n A_i = \frac{8}{3}(32)^2$ (D) $\lim_{n \rightarrow \infty} \sum_{i=1}^n A_i = \frac{4}{3}(16)^2$
12. Let T be the triangle with vertices $(0, 0)$, $(0, c^2)$ and (c, c^2) and let R be the region between $y = cx$ and $y = x^2$ where $c > 0$ then
- (A) $\text{Area}(R) = \frac{c^3}{6}$ (B) $\text{Area of } R = \frac{c^3}{3}$
- (C) $\lim_{c \rightarrow 0^+} \frac{\text{Area}(T)}{\text{Area}(R)} = 3$ (D) $\lim_{c \rightarrow 0^+} \frac{\text{Area}(T)}{\text{Area}(R)} = \frac{3}{2}$

Comprehension Type Question:

Comprehension # 1

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous function and bijective, defined such that $f(\alpha) = 0$ ($\alpha \neq 0$). The area bounded by $y = f(x)$, $x = \alpha$, $x = \alpha - t$ is equal to area bounded by $y = f(x)$, $x = \alpha$, $x = \alpha + t$, $\forall t \in \mathbb{R}$ then

13. $y = f(x)$ is symmetrical about the point
- (A) $(0, 0)$ (B) $(0, \alpha)$ (C) $(\alpha, 0)$ (D) (α, α)
14. $f(2\alpha)$ equal to ____.
- (A) $f(\alpha)$ (B) $-f(\alpha)$ (C) $f(0)$ (D) $-f(0)$

Comprehension # 2

If $y = f(x)$ is a monotonic function in (a, b) , then the area bounded by the ordinates at $x = a$, $x = b$, $y = f(x)$ and $y = f(c)$ (where $c \in (a, b)$) is minimum when $c = \frac{a+b}{2}$.

$$\begin{aligned} \text{Proof: } A &= \int_a^c (f(c) - f(x)) dx + \int_c^b (f(x) - f(c)) dx \\ &= f(c)(c-a) - \int_a^c f(x) dx + \int_c^b f(x) dx - f(c)(b-c) \\ \Rightarrow A &= [2c - (a+b)f(c) + \int_c^b f(x) dx - \int_a^c f(x) dx] \end{aligned}$$

Differentiating w.r.t. c ,

$$\frac{dA}{dc} = [2c - (a+b)] f'(c) + 2f(c) + 0 - f(c) - (f(c))$$

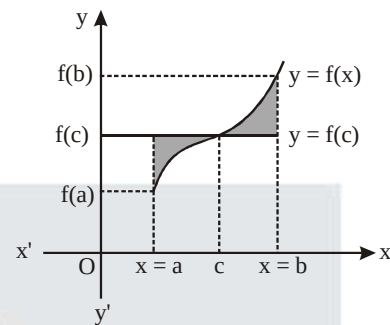
for maxima and minima $\frac{dA}{dc} = 0$

$$\Rightarrow f'(c)[2c - (a+b)] = 0 \text{ (as } f'(c) \neq 0 \text{)}$$

$$\text{hence } c = \frac{a+b}{2}$$

$$\text{Also } c < \frac{a+b}{2}, \frac{dA}{dc} < 0 \text{ and } c > \frac{a+b}{2}, \frac{dA}{dc} > 0.$$

Hence A is minimum when $c = \frac{a+b}{2}$.



15. If the area bounded by $f(x) = \frac{x^3}{3}$ and the straight lines $x = 0$, $x = 2$ and the x -axis is minimum, then the value of a is
 (A) $\frac{1}{3}$ (B) 2 (C) 1 (D) $\frac{2}{3}$
16. The value of the parameter a for which the area of the figure bounded by the abscissa axis, the graph of the function $y = x^3 + 3x^2 + x + a$ and the straight lines, which are parallel to the axis of ordinates and cut the abscissa axis at the point of extremum of the function, is the least, is
 (A) 2 (B) 0 (C) -1 (D) 1
17. If the area enclosed by $f(x) = \sin x + \cos x$, $y = a$ between the two consecutive points of extremum is minimum, then the value of a is
 (A) 0 (B) -1 (C) 1 (D) 2

Comprehension # 3

Consider the areas $S_0, S_1, S_2, \dots$ bounded by the x -axis and half-waves of the curve $y = e^{-x} \sin x$, where $x \geq 0$.

18. The value of S_0 is

- (A) $\frac{1}{2}(1 + e^\pi)$ sq. units (B) $\frac{1}{2}(1 + e^{-\pi})$ sq. units
 (C) $\frac{1}{2}(1 - e^{-\pi})$ sq. units (D) $\frac{1}{2}(e^\pi - 1)$ sq. units.

19. The sequence $S_0, S_1, S_2, \dots$, forms a G.P. with common ratio

- (A) $\frac{e^\pi}{2}$ (B) $e^{-\pi}$ (C) e^π (D) $\frac{e^{-\pi}}{2}$

20. $\sum_{n=0}^{\infty} S_n$ is equal to

- (A) $\frac{1+e^\pi}{1-e^{-\pi}}$ (B) $\frac{1}{2} \frac{(1+e^\pi)}{1-e^{-\pi}}$ (C) $\frac{1}{2(1-e^{-\pi})}$ (D) None of these

Numerical based Questions :

21. If A is the area of the figure bounded by the straight line $x = 0$ and $x = 2$ and the curves $y = 2^x$ and $y = 2x - x^2$, then the value of $6\left(\frac{3}{\ln 2} - A\right)$ is
22. The graph of $y^2 + 2xy + 40|x| = 400$ divides the plane into regions. The area of the bounded region is
23. Area enclosed by the curve $4 \leq x^2 + y^2 \leq 2(|x| + |y|)$ is (in sq. units)
24. The area bounded by the curves $y = -\sqrt{4-x^2}$, $x^2 = -y\sqrt{2}$ and $x = y$ can be expressed in the form of $\pi + \frac{\lambda}{\mu}$, (λ, μ being relatively prime positive integers) then $\lambda + \mu$ equals to _____.
25. Find the area enclosed between the curves : $y = \log_e(x+e)$, $x = \log_e(1/y)$ & the x-axis.

Matrix Match Type :

26. Match the following

Column-I	Column-II
(a) Let $f(K) = \frac{K}{2009}$ and $g(K) = \frac{(f(K))^4}{(1-f(K))^4 + (f(K))^4}$ and $S = \sum_{K=0}^{2009} g(K)$ then sum of the diagonal in S =	(p) 0
(b) $f : D \rightarrow \mathbb{R}$ such that $f(k) = \sqrt{\sin(\cos x)} + \log_e(-2\cos^2 x + 3\cos x + 1)$ then $\int_{x_1}^{x_2} \left[\cos x - \frac{1}{2} \right] dx$ is equal to, $x_1, x_2 \in D$ and $[.]$ denotes greatest integer function.	(q) 1
(c) The area of region, represented by $ x + y \leq 2$ and $x^2 \leq y$ is S square units then $3S =$ _____.	(r) 7
(d) $\lim_{y \rightarrow \infty} \frac{\int_1^y [\tan^{-1} x] dx}{\int_1^y \left[1 + \frac{1}{x} \right] dx} =$ _____. ($[.]$ denotes greatest integer function)	(s) 6

- (A) $A \rightarrow s$; $B \rightarrow p$; $C \rightarrow s$; $D \rightarrow q$ (B) $A \rightarrow p$; $B \rightarrow s$; $C \rightarrow r$; $D \rightarrow q$
 (C) $A \rightarrow s$; $B \rightarrow p$; $C \rightarrow r$; $D \rightarrow q$ (D) $A \rightarrow s$; $B \rightarrow p$; $C \rightarrow r$; $D \rightarrow s$

27. Column - I

Column-II

(A) The area bounded by the curve $y = x|x|$, x-axis and the ordinates $x = 1$, $x = -1$

(P) $\frac{10}{3}$ sq. units

(B) The area of the region lying between the line $x - y + 2 = 0$ and the curve $x = \sqrt{y}$

(Q) $\frac{64}{3}$ sq. units

(C) The area enclosed between the curves $y^2 = x$ and $y = |x|$

(R) $\frac{2}{3}$ sq. units

(D) The area bounded by parabola $y^2 = x$, straight line $y = 4$ and y-axis

(S) $\frac{1}{6}$ sq. units

(A) $A \rightarrow Q$; $B \rightarrow S$; $C \rightarrow P$; $D \rightarrow R$

(B) $A \rightarrow S$; $B \rightarrow R$; $C \rightarrow Q$; $D \rightarrow P$

(C) $A \rightarrow R$; $B \rightarrow P$; $C \rightarrow S$; $D \rightarrow Q$

(D) $A \rightarrow Q$; $B \rightarrow R$; $C \rightarrow P$; $D \rightarrow S$

